

Function Code(Optional)	Sheet Name	Service Name	Description	Pre-Condition
register	register	Authentication Service	Registers a new user account with email, password, and full name; issues JWT tokens on success	Email must not be already registered; MEMBER role must exist in the database
login	login	Authentication Service	Authenticates a user with email and password; returns access and refresh tokens	User must exist; account status must be ACTIVE (not INACTIVE or SUSPENDED)
me	me	Authentication Service	Returns the authenticated user's profile information	User authenticated (valid Bearer access token); account must be ACTIVE
refresh	refresh	Authentication Service	Issues a new access/refresh token pair using a valid unexpired refresh token	Valid refresh token must be provided (body or cookie); token must not be expired or revoked; account must be ACTIVE
logout	logout	Authentication Service	Revokes the current refresh token and ends the user's session	User authenticated; optional refresh token provided to revoke specific session
listUsers	listUsers	User Management Service	Returns a paginated list of all users in the system	User authenticated; ADMIN role required
getUserById	getUserById	User Management Service	Retrieves a single user's public profile by their ID	User authenticated
updateUser	updateUser	User Management Service	Updates a user's profile fields (full name, avatar, bio)	User authenticated; requester must be the same user (own profile)

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setUserStatus	setUserStatus	User Management Service	Changes a user's account status (ACTIVE / INACTIVE / SUSPENDED)	User authenticated; ADMIN role required
removeUser	removeUser	User Management Service	Soft-deactivates a user account (marks as INACTIVE)	User authenticated; ADMIN role required
listGroups	listGroups	Study Group Service	Returns a paginated, filterable list of study groups; hides non-ACTIVE groups from non-admins	Public (optional authentication); ADMIN users see all statuses
getGroupById	getGroupById	Study Group Service	Returns details of a specific study group including member count and join request status	Public (optional authentication)
createGroup	createGroup	Study Group Service	Creates a new study group and automatically adds the creator as a LEADER member	User authenticated
updateGroup	updateGroup	Study Group Service	Updates group metadata (name, description, subject, maxMembers)	User authenticated; requester must be the group LEADER or ADMIN

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setGroupStatus	setGroupStatus	Study Group Service	Changes a group's lifecycle status (ACTIVE / ARCHIVED / DELETED)	User authenticated; ADMIN role required
removeGroup	removeGroup	Study Group Service	Soft-deletes a study group	User authenticated; requester must be group LEADER or ADMIN
requestJoin	requestJoin	Study Group Service	Submits a join request to a study group	User authenticated; user must not already be a member or have a PENDING request
cancelJoinRequest	cancelJoinRequest	Study Group Service	Cancels the authenticated user's pending join request to a group	User authenticated; PENDING join request must exist for the user and group
approveJoinRequest	approveJoinRequest	Study Group Service	Approves a pending member join request, adding the user as a MEMBER	User authenticated; requester must be group LEADER; join request must be PENDING
rejectJoinRequest	rejectJoinRequest	Study Group Service	Rejects a pending member join request	User authenticated; requester must be group LEADER; join request must be PENDING
handleJoinRequest	handleJoinRequest	Study Group Service	Generic handler to approve or reject a join request by setting its status	User authenticated; requester must be group LEADER; join request must exist
listSessions	listSessions	Study Session Service	Returns a paginated, filterable list of sessions (by group, status, upcoming, or own sessions)	Public (optional authentication)

Function Code(Optional)	Sheet Name	Service Name	Description	Pre-Condition
getSessionById	getSessionById	Study Session Service	Returns details of a specific study session	Public (optional authentication)
createSession	createSession	Study Session Service	Creates a new study session (SCHEDULED or immediately IN_PROGRESS); optionally notifies group members	User authenticated; user must be a group member
updateSession	updateSession	Study Session Service	Updates a session's metadata (title, description, time, meeting link)	User authenticated; requester must be session creator or group LEADER
endSession	endSession	Study Session Service	Marks a session as COMPLETED (if live) or CANCELLED; deletes the LiveKit room if applicable	User authenticated; session must not already be COMPLETED or CANCELLED; requester must be creator or LEADER
notifyMembers	notifyMembers	Study Session Service	Sends an in-app notification to all group members (except the sender) about a session	User authenticated; session must exist; requester must be creator or LEADER
getLiveKitToken	getLiveKitToken	Study Session Service	Generates a LiveKit JWT room access token for the authenticated user to join the live session video room	User authenticated; session must be IN_PROGRESS; user must be a group member; LiveKit must be configured

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removeSession	removeSession	Study Session Service	Soft-deletes a session and removes its LiveKit room	User authenticated; requester must be the session creator
markAttendance	markAttendance	Attendance Service	Records or updates a user's attendance status (PRESENT / ABSENT / LATE / EXCUSED) for a session	User authenticated; session must exist; requester must be session creator or group LEADER to mark others
recordJoin	recordJoin	Attendance Service	Automatically marks a user as PRESENT when they join a LiveKit session room	User authenticated; session must exist
listAttendance	listAttendance	Attendance Service	Returns the full attendance record list for a given session	User authenticated
listResourceFolders	listResourceFolders	Resource Folder Service	Returns all resource folders for a given group	User authenticated
getResourceFolderById	getResourceFolderById	Resource Folder Service	Returns details of a specific resource folder	User authenticated
createResourceFolder	createResourceFolder	Resource Folder Service	Creates a new named resource folder within a study group	User authenticated; user must be a group member
updateResourceFolder	updateResourceFolder	Resource Folder Service	Updates a resource folder's name or description	User authenticated; requester must be the folder creator or group LEADER

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listResources	listResources	Resource Management Service	Returns a paginated list of resources, filterable by group or folder	User authenticated
getResourceById	getResourceById	Resource Management Service	Returns details of a specific resource including star count	User authenticated
createResource	createResource	Resource Management Service	Adds a new resource (file URL, title, type, folder) to a group	User authenticated; user must be a group member folder must exist in the same group
toggleStarResource	toggleStarResource	Resource Management Service	Toggles the authenticated user's star rating on a resource (adds or removes)	User authenticated; resource must exist
removeResource	removeResource	Resource Management Service	Soft-deletes a resource	User authenticated; requester must be the resource uploader or group LEADER
listPosts	listPosts	Discussion Post Service	Returns a paginated, filterable list of group discussion posts	User authenticated
getPostById	getPostById	Discussion Post Service	Returns a single post with its vote score and user's own vote	User authenticated
createPost	createPost	Discussion Post Service	Creates a new discussion post in a study group	User authenticated; user must be a group member

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updatePost	updatePost	Discussion Post Service	Edits a post's title, content, or attachments	User authenticated; requester must be the post author
votePost	votePost	Discussion Post Service	Records or updates the user's vote (+1 upvote / -1 downvote / 0 removes) on a post	User authenticated; post must exist
removePost	removePost	Discussion Post Service	Soft-deletes a discussion post	User authenticated; requester must be the post author or group LEADER/ADMIN
listCommentsByPost	listCommentsByPost	Comment Service	Returns a paginated, threaded list of comments for a specific post	User authenticated; post must exist
createComment	createComment	Comment Service	Adds a new comment (or threaded reply) to a post, with optional user mentions	User authenticated; post must exist; parent comment must exist if replying
updateComment	updateComment	Comment Service	Edits a comment's content	User authenticated; requester must be the comment author
voteComment	voteComment	Comment Service	Records or updates the user's vote on a comment (+1 / -1 / 0)	User authenticated; comment must exist
removeComment	removeComment	Comment Service	Soft-deletes a comment	User authenticated; requester must be the comment author or group LEADER/ADMIN

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listNotifications	listNotifications	Notification Service	Returns a paginated list of the authenticated user's notifications, with optional unread filter	User authenticated
getUnreadCount	getUnreadCount	Notification Service	Returns the count of unread notifications for the authenticated user	User authenticated
markNotificationRead	markNotificationRead	Notification Service	Marks a single notification as read	User authenticated; notification must belong to the requesting user
markAllNotificationsRead	markAllNotificationsRead	Notification Service	Marks all of the user's notifications as read	User authenticated
getStats	getStats	Dashboard & Analytics Service	Returns role-aware aggregated statistics: ADMIN sees platform-wide totals; LEADER sees their group summaries; MEMBER sees their personal activity	User authenticated
getGroupStats	getGroupStats	Dashboard & Analytics Service	Returns detailed statistics for a specific group (member count, session count, attendance rate, resource count)	User authenticated; ADMIN role required; group must exist

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getCloudinarySignature	getCloudinarySignature	File Upload Service	Generates a signed Cloudinary upload credential (timestamp + signature) so the client can upload files directly to Cloudinary without exposing the API secret	User authenticated; Cloudinary must be configured on the server (CLOUDINARY_CLOUD_NAME, CLOUDINARY_API_KEY, CLOUDINARY_API_SECRET) environment variables set